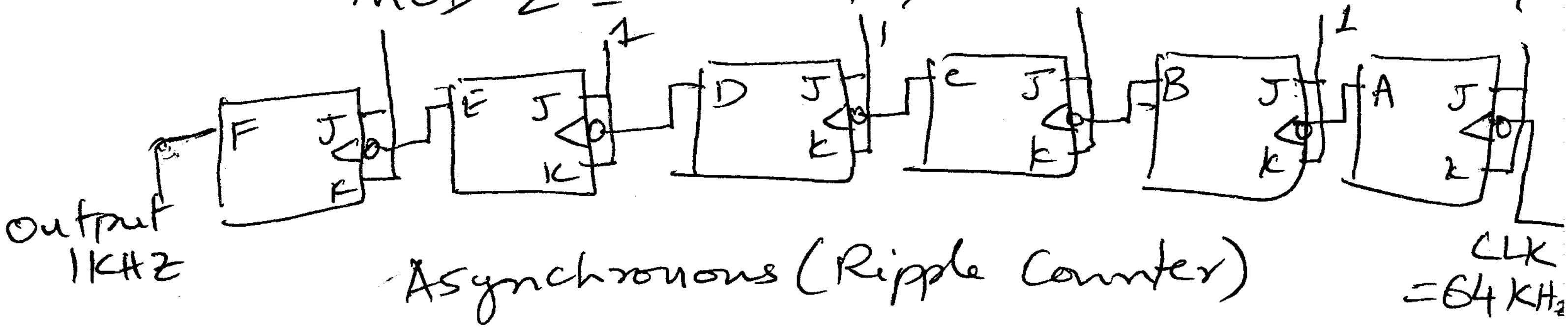
HW - CH 7

7.2 Need to divide by 64; use MOD-64, 6-bit counter

64 KHZ ÷ 64 = 1 KHZ So use MOD 64 ⇒
MOD 2^6 = MOD 64, So use 6 FF



Asynchronous (Ripple Counter)

7.4 (a) 1024

10 bits means counting is upto 2^{10}

$2^{10} = 1024$

So MOD 1024

(b) 250 Hz

Frequency of MSB output is $\frac{\text{INPUT CLK}}{1024}$

$= \frac{256 \times 10^3}{1024} = 250\,Hz$

(c) 50%

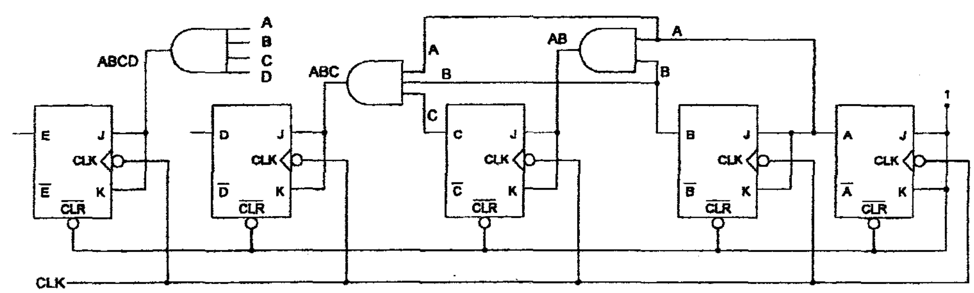
Since output is Square wave
Duty cycle = 50%.

(d) 3E8

$$16 \,|\, 1000 \quad 8$$
$$16 \,|\, 62 \quad 14\ (E)$$
$$3$$

3E8 is the count in Hex

7.7 (a) See schematic (b) 33 MHz



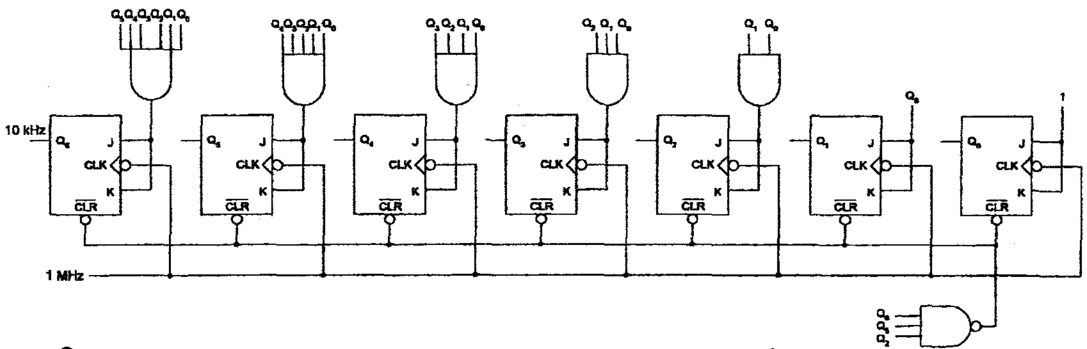
t_{pd} of FF = 20 ns

t_{pd} of gate = 10 ns

$T_{clock} \geq 20 + 10$ ns $= 30$ ns

$f_{max} = \frac{1}{30\,ns} = 33$ MHZ

7.12



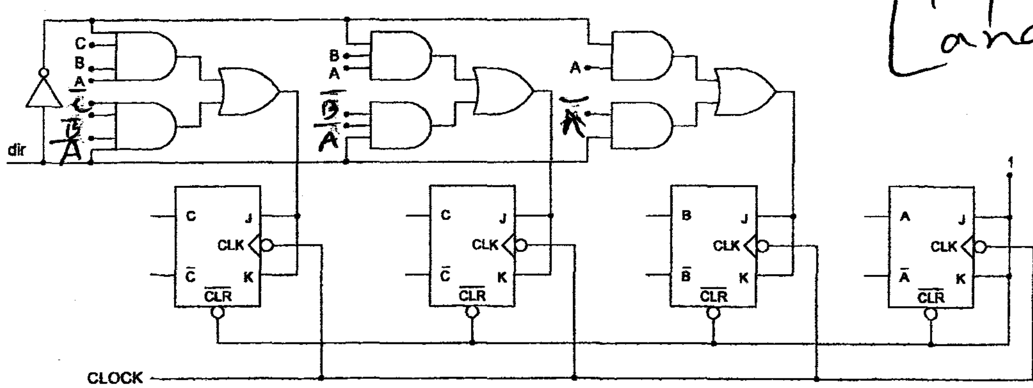
$f_{output} = 10\ \text{KHZ} \qquad CLK = 1\ \text{MHz}$

$$\text{MOD}\ \# = \frac{\text{CLK freq}}{f\ \text{output}} = 100$$

So MOD 100 counter is needed $2^6 = 64$ & $100 < 2^7 \rightarrow$ so 7 flip flops are needed

Binary of 100 is $\underset{Q_7}{0}\,\underset{Q_6}{1}\,\underset{Q_5}{1}\,\underset{Q_4}{0}\,\underset{Q_3}{0}\,\underset{Q_2}{1}\,\underset{Q_1}{0}\,\underset{Q_0}{0} \Rightarrow$ Use a NAND gate whose inputs are Q_2, Q_5, and Q_7

7.14



Note when dir = 0, the AND gates that gets connected to A, AB, ABC is activated because $\overline{\text{dir}} = 1$. The signal that gets connected to these gates are inverted dir. Please understand the other case ie dir = 1